Practice Set — Unit II

Concepts in Metric Spaces

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Dr. Lokesh Singh
Department of Mathematics, Amity University Punjab, Mohali

Questions

1. **(Proof — Metric Axioms)** Let X be any nonempty set. Define

$$d(x,y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases}$$

   Prove carefully that d satisfies all three metric axioms, and hence that (X, d) is a metric space.

2. **(Computation)** Let $x = (1, 3)$ and $y = (5, 0)$ in $\mathbb{R}^2$. Compute:

$$\text{(a) } d_1(x,y), \quad \text{(b) } d_2(x,y), \quad \text{(c) } d_\infty(x,y).$$

   Rank the three values and explain in one sentence why this ranking always holds for any two points in $\mathbb{R}^2$.

3. **(Visualization)** In $\mathbb{R}^2$, sketch and clearly label the unit ball $B_1(0)$ for each of the following metrics:

$$d_1, \qquad d_2, \qquad d_\infty.$$

   Then determine for each metric whether the point $P = (0.7, 0.6)$ belongs to $B_1(0)$. Justify your answer.

4. **(Computation)** Let $X = \{a, b, c\}$ with the discrete metric.

   (a) Find $B_1(a)$, $B_{1/2}(a)$, and $B_2(a)$.

(b) Find the sphere $S_1(a)$ and the closed ball $\overline{B}_1(a)$.

(c) Identify all interior points, all isolated points, and all limit points of the set $\{a, b\}$.

5. **(Proof)** Let (X, d) be a metric space, $x_0 \in X$, and $r > 0$. Prove that the closed ball

$$\overline{B}_r(x_0) = \{y \in X : d(x_0, y) \leq r\}$$

is a closed subset of X.

6. **(Proof)** Let (X, d) be a metric space and $x, y, z \in X$. Prove that

$$\big|d(x, z) - d(y, z)\big| \leq d(x, y).$$

Then explain in one sentence how this result shows that the function $f : X \to \mathbb{R}$ defined by $f(z) = d(x, z)$ is 1-Lipschitz.

7. **(Computation)** In $\mathbb{R}$ with the usual metric, find $\text{int}(A)$, $\overline{A}$, and ∂A for each set below. Briefly justify each answer.

(a) $A = (0, 1) \cup (1, 2)$

(b) $A = [0, 1) \cup \{3\}$

(c) $A = \mathbb{Q} \cap [0, 1]$

8. **(Proof)** Let (X, d) be a metric space.

(a) Prove that the union of any collection of open sets is open.

(b) Prove that the intersection of two open sets is open.

(c) Give an explicit example in $\mathbb{R}$ showing that the intersection of infinitely many open sets need not be open.

9. **(Visualization and Classification)** In $\mathbb{R}^2$ with the Euclidean metric, sketch the following sets and classify each as **open**, **closed**, **neither**, or **both**. Justify each classification.

(a) $A = \{(x, y) : x^2 + y^2 < 1\}$

(b) $B = \{(x, y) : x^2 + y^2 = 1\}$

(c) $C = \{(x, y) : x > 0,\ y > 0\}$

(d) $D = \{(x, y) : x \geq 0,\ y > 0\}$

10. **(Computation and Concept)** In $\mathbb{R}$ with the usual metric, find all limit points and all isolated points of each set.

(a) $A = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$

(b) $B = \mathbb{Z}$

(c) $C = \mathbb{Q} \cap (0, 1)$

For each set, state whether it is closed, and explain your reasoning in one line.

11. **(Proof)** Let X be a nonempty set with the discrete metric. Prove that:

(a) Every subset of X is open.

(b) Every subset of X is closed.

(c) X has no limit points.

12. **(Computation and Proof)**

(a) In $\mathbb{R}$ with the usual metric, compute $\operatorname{diam}(A)$ for $A = (0,1)$, $A = [0,1]$, and $A = \mathbb{Q} \cap [0,1]$.

(b) In the discrete metric space on any set X, compute $\operatorname{diam}(A)$ for any set A with $|A| \geq 2$.

(c) Prove that if $A \subseteq X$ is bounded and $B \subseteq A$, then B is also bounded.

13. **(Computation)** Let $Y = [0,1]$ be a subspace of $\mathbb{R}$ (with the usual metric).

(a) Is $[0, 1/2)$ relatively open in Y? Justify.

(b) Is $[0, 1/2]$ relatively closed in Y? Justify.

(c) Is $(1/3, 2/3]$ relatively open in Y? Justify.

(d) Is Y itself clopen in Y? Explain in one line.

14. **(Proof and Concept)** Let $X = \mathbb{R}$, and define on X:

$$d_1(x,y) = |x-y|, \qquad d_2(x,y) = \min\{1, |x-y|\}.$$

(a) Verify that d_2 is a metric on $\mathbb{R}$.

(b) Show that d_1 and d_2 are **topologically equivalent**, i.e., they generate the same collection of open sets.

(c) Show that d_1 and d_2 are **not Lipschitz equivalent** (so the stronger equivalence fails).

(d) Give one property that is preserved by $d_1 \leftrightarrow d_2$ and one property that is **not** preserved. Justify.

15. **(Proof — Extension)**

(a) On $\mathbb{R}^n$, prove the chain of inequalities

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1 \leq n\,\|x\|_\infty,$$

where $\|x\|_p = \left(\sum_{k=1}^n |x_k|^p\right)^{1/p}$ for $p \in \{1,2\}$ and $\|x\|_\infty = \max_k |x_k|$.

(b) Conclude that d_1, d_2, d_∞ are **Lipschitz equivalent** on $\mathbb{R}^n$.

(c) Recall that ℓ^p denotes the space of sequences (x_n) with $\sum_n |x_n|^p < \infty$. Show that

$$\ell^1 \subsetneq \ell^2 \subsetneq \ell^\infty.$$